Jekaterina Novikova, PhD

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Google Scholar
Personal website

Research in Natural Language Processing and Machine Learning. Author of 60+ peer reviewed publications (EMNLP, ACL, SigDIAL etc.) and 3 patent applications. Organized the End-to-End NLG shared task which set a new research agenda for neural generation, and co-led the development of the BLOOM large language model. Experience in building and leading successful research teams.

EXPERIENCE

Director of Machine Learning Research

Cambridge Cognition (prev. Winterlight Labs), Toronto/Canada

May 2018 - PRESENT

- Setting up a research agenda based on the SOTA work in ML and NLP.
- Building and leading a diverse team of ML research scientists and engineers of various seniority.
- Driving cross functional collaborations within the company and with external collaborators, e.g. co-led the development of the BLOOM large language model.
- Influencing the general public as well as the community of AI practitioners via presentations and panel discussions at local and international events, such as ODSC and MLconf conferences, Re-Work AI symposium, Health Data Law conference, meet-ups and other venues.

Science Lead

AI Risk and Vulnerability Alliance / AVID, non-profit, USA

December 2022 - PRESENT

- Setting up a research agenda based on the SOTA work in LLMs risks and harms.
- Building a community of ML researchers interested in the topic of AI safety, risks and harms.
- Building out a functional taxonomy of potential AI harms across the coordinates of security, ethics, and performance.

Postdoc Researcher

Heriot-Watt University, Edinburgh/UK

November 2015 - April 2018

Organizing a successful shared challenge "E2E NLG Challenge":

- Collected the largest at the time dataset for the NLG research, inspired collection of other new datasets and further academic work in the area.
- Organized the End-to-End NLG shared task which set a new research agenda for neural generation.
- Attracted participants from all over the world.
- Developed evaluation strategies for language-producing models.
- Presented the challenge, associated research work, and the results at academic conferences.

SKILLS

Publications:

EMNLP, ACL*, Interspeech etc.
h-index 18
2800+ citations

Technical:

Python, PyTorch, W&B, Git, GitHub

Public speaking:

TMLS 2023 / Toronto, Canada

BigScience 2021 / Virtual

ODSC 2020 / Virtual
MLconf 2019 / SF, USA

ODSC 2019 / Boston, USA

Re-Work AI Summit 2018 / Toronto, Canada

CogX 2017 / London, UK

AWARDS

Top 25 Women in AI in Canada
by Re-Work, 2022

Industry Icon Award
by University of Toronto, 2021

30 Influential Women Advancing AI in Canada
by Re-Work, 2018

Best paper nomination:
SIGDIAL 2017
HAI 2014

Developing a conversational system for a humanoid robot Pepper allowing it to communicate with people in a shopping mall:

- Participated in a EU-wide collaborative project with 7 partner institutions, including Softbank Robotics, Universities and Research Centres.
- Conducted research related to developing a conversational system combining a variety of interaction modalities, such as speech, graphics, gestures, vision, and natural language.
- Collaboratively developed the app for the robot's installation into a food store. The experiment was filmed by the BBC for the *Six Robots & Us* episodes.
- Presented demos of the robot to the general public at e.g. Edinburgh Science Festival.
- Presented the project, associated research work, and the results at the academic conferences.

Visiting Researcher

National Institute of Science, Tokyo/JP

October 2014 - December 2014

- Modeled a human-robot interaction task in a SIGVerse simulator (C++).
- Prepared and performed experiments with human subjects interacting with a virtual robot.
- Presented research results at conferences (HRI, Ro-Man).

EDUCATION

PhD at the University of Bath, UK, 2012 - 2015

MSc CompSci at Blekinge Institute of Tech, Sweden, 2009 - 2011

MSc Business Informatics and
BA Business Management(major) and CompSci(minor)
at Vytautas Magnus University, Lithuania, 1996 - 2004

SELECTED PUBLICATIONS

- 300+ authors incl. J. **Novikova** (2022). BLOOM: A 176B-Parameter Open-Access Multilingual Language Model. *arXiv:2211.05100* [total 588 citations as of October 2023]
- 400+ authors incl. J. **Novikova** (2023). Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models, in *Journal of Machine Learning Research*, Vol. 24. [Impact factor: 5.1]
- A. Balagopalan, B. Eyre, J. Robin, F. Rudzicz, J. **Novikova** (2021). Comparing Pre-trained and Feature-based Models for Prediction of Alzheimer's Disease Based on Speech, in *Frontiers in Aging Neuroscience*, Vol. 13, p. 189 [Impact factor: 4.8]
- A. Balagopalan, B. Eyre, F. Rudzicz, J. **Novikova** (2020). To BERT or Not To BERT: Comparing Speech and Language-based Approaches for Alzheimer's Disease Detection. In: *INTERSPEECH 2020*
- Z. Zhu, J. **Novikova**, and F. Rudzicz (2019). Detecting Cognitive Impairments

Best talk award:
Afternoon of research 2014, Uni of Bath

COMMUNITY INVOLVEMENT

Chair:
GEM at EMNLP 2023, GEM at EMNLP 2021

Reviewer:

EMNLP, ACL, NAACL, AACL, NeurIPS, AAAI, ICLR, ICDL/EpiRob, CHI, ICMI, HAI, TAROS, conferences

Associate editor:

Journal of Adaptive Behavior

Organizing committee:

ML4CMH Workshop

Gem Workshop

E2E NLG Shared Challenge

INLG 2016 Conference

LANGUAGES

English, Russian, Lithuanian

by Agreeing on Interpretations of Linguistic Features. In *Proceedings of NAACL 2019* (22% acceptance rate)

- O. Dusek, J. **Novikova**, V. Rieser (2019). Evaluating the State-of-the-Art of End-to-End Natural Language Generation. In *Journal of Computer Speech and Language,* Vol. 59, pp. 123-156 [Impact factor: 4.3]
- J. **Novikova**, O. Dusek and V. Rieser (2018). RankME: Reliable Human Ratings for Natural Language Generation. In *Proceedings of NAACL 2018* (29% acceptance rate)
- J. **Novikova**, O. Dusek, A. Cercas-Curry and V. Rieser (2017): Why We Need New Evaluation Metrics for NLG. In *Proceedings of EMNLP 2017* (22% acceptance rate)
- J **Novikova**, O. Dusek and V. Rieser (2017). The E2E Dataset: New Challenges For End-to-End Generation. In *Proceedings of SIGDIAL 2017*

OTHER PROFESSIONAL ACTIVITIES

- Expert Advisor - ACL Special Interest Group SIGSEC (Aug 2023 - Present)
- Steering committee member - Toronto Machine Learning Society (July 2020 - Present)
- Mentor - Interstellar Initiative, collaboration between Japan Agency for Medical Research and Development and the New York Academy of Sciences (July 2020 - Feb 2023)
- Program Committee member and/or Area Chair - EMNLP, ACL*, ARR, NeurIPS, ICLR, AAAI

SELECTED INVITED TALKS

Toronto Machine Learning Summit. Paving the Road to Trustworthy AI: Bias in Large Language Models. Toronto, June 2023

TMLS Summit on NLP. Interpretability and Robustness of Transformer Models in Healthcare. Toronto, September 2022

BigScience - Evaluation. Evaluation of Model Robustness and Consistency. Virtual, October, 2021

ODSC West, Ai x Summit. Solving Real-life Challenges in Detecting Cognitive Diseases from Speech Using ML. October 2020

CAMDEA Digital Forum - Conversational AI, Chatbot and Voice Technology. November 2020

MLconf. Machine Learning Methods in Detecting Alzheimer's Disease from Speech and Language. San Francisco, November 2019 [video]

ODSC East, Women in Data Science Ignite. Classification of Alzheimer's Disease from Speech and Language. Boston, May 2019

Re-Work, Women in AI. Language and Speech Processing: From Human-Robot Interaction to Alzheimer's Prediction. Toronto, October 2018

CogX London. Socially Intelligent Robot and Its Impact on Customer Experience in a Shopping Mall. London/UK, June 2017 [video]

Re-Work Deep Learning in Retail & Advertising Summit. Socially Intelligent Robot in a Shopping Mall. London/UK, June 2017